

GETTIN' GRIDDY WITH IT: OPTIMAL SEISMIC PROCESSING AND MODELING OF THE YAMPA BASIN IN STEAMBOAT SPRINGS, COLORADO

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ABSTRACT

The Yampa River Basin near Steamboat Springs, Colorado, preserves a complex record of basement faulting, sedimentary basin development, and Quaternary surface processes, but its subsurface structure remains poorly constrained. The region's strong lateral and vertical heterogeneity complicates seismic processing, adding challenges to the development of reliable seismic sections and subsurface models. We acquired 13 km of active-source seismic data using a vibroseis truck along Routt County Roads (RCR) 14 and 22, south of Steamboat Springs. These data were integrated with existing seismic data along Colorado Highway 131. To resolve complex subsurface geology, we used SeisSpace and Petrel to generate 2D sections and a gridded model. To image the basin's subsurface, we applied refraction and residual statics to correct for near-surface and localized heterogeneities. Normal moveout corrections were applied based on interpolations between several velocity models along the line. The sections were then converted to depth with turning-ray velocity models and processed using a Kirchhoff migration algorithm. The processed models were visualized using Petrel seismic software. The migrated cross sections were connected into a 3-dimensional grid, which was used to create a depth-structure map. Applying the most optimal processing through various corrections and processing methods, we were able to obtain accurate seismic sections and models of the Yampa river basin from these data. These optimized methods provide a valuable framework to inform future geophysical investigations in the Steamboat Springs region and geologically analogous environments. The produced model will also inform analysis and interpretation of the basin's complex and understudied geology. For detailed geological interpretations of these processed sections, including structural analysis of the Precambrian basement, see the companion abstract by Larner et al.